UNIVERSITY OF MALAYA

EXAMINATION FOR THE DEGREE OF MASTER OF DATA SCIENCE

ACADEMIC SESSION 2017/2018 : SEMESTER II

WQD7006 : Machine Learning for Data Science

May/June 2018 Time : 3 hours

INSTRUCTIONS TO CANDIDATES :

Answer **ALL** questions (50 marks).

(This question paper consists of 5 questions on 5 printed pages)

1.

a. State the main differences between supervised and unsupervised learning?

(2 marks)

b. Can you predict continuous values using a classification based learning algorithm? Justify your answer.

(2 marks)

c. We have learnt two different types of unsupervised learning method. State a specific unsupervised problem for each of the method, and discuss briefly how the method can be used to solve each of the problem.

(6 marks)

2.

a. Define ensemble methods within the context of Machine Learning. Name two ensemble methods and describe in brief how it works.

(6 marks)

b. With reference to classifier ensembles (meta-learning), classify which ones of the following statements are true (multiple choice):

(1) Classifier ensembles usually have better performance for weak classifiers because they combine the different points of view of each individual classifier

(2) Boosting is an ensemble method that uses sampling with replacement of a dataset and learns independently several classifiers

(3) Random subspaces methods (like random forest) are meta-learning methods that learn a set of classifiers using datasets that have subsets of attributes of the original data

(4) For methods like Boosting or Bagging to perform well, it is necessary that the different subsamples from the dataset used in each classifier are as similar as possible

(4 marks)

3. Figure 1 is a representation of one unit of artificial neuron, i.e. a node (with no bias). The following questions are based on Figure 1.

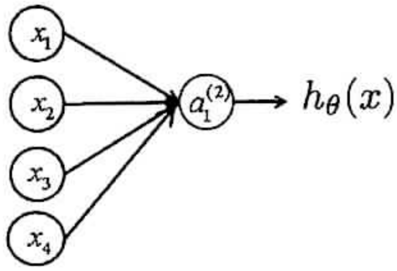

Figure 1

a. The node has four inputs $x = x_1, x_2, x_3, x_4$ that receive only binary signals (either 0 or 1). Compute how many different input patterns this node can receive?

(2 marks)

b. What if the node has five inputs? Six? Can you formulate a formula that computes the number of binary input patterns for a given number of inputs?

(2 marks)

c. Suppose that the weights corresponding to the three inputs have the following values:

$$w_1 = 2$$
$$w_2 = -4$$
$$w_3 = 1$$
$$w_4 = 3$$

and the activation of the unit is given by the step-function:

$$a_i^{(j)} = \varphi(a_i^{(j)}) = \begin{cases} 1 & \text{if } a_i^{(j)} \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Note: $a_i^{(j)} =$ "Activation" of unit *i* in layer *j*

Calculate what will be the output value, $h_\theta(x)$ of the unit for each of the following input patterns:

Pattern, P_i	P_1	P_2	P_3
X_1	1	0	1
X_2	0	1	0
X_3	0	1	1
X_4	1	1	1

(6 marks)

4.

a. State the name of the algorithm used when fitting a Gaussian Mixture Model.
(1 marks)

b. The algorithm for fitting Gaussian Mixture Models indirectly optimizes the likelihood of the data by instead optimizing the free energy. Explain the relationship between the free energy and the likelihood of the data?
(2 marks)

c. For the K-Means algorithm: State what are the inputs? State which parameters are usually specified by the user and what does the algorithm estimate?

(3 marks)

d. Specify and explain (briefly) the steps of the K-Means algorithm.

(4 marks)

5. Principal component analysis (PCA) is a statistical method for identifying patterns in data. It can also be used to compress data via dimensionality reduction. Specify and explain the different steps needed to perform PCA on a set of data provided in the Table 1.

You **do not need** to compute the eigenvalues and eigenvectors for this question. However, **other** steps of PCA that requires mathematical computations to support your explanation shall be taken into consideration during the marking of your answer to this question.

Table 1: Data for PCA

x	y
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
3.1	3.0
2.3	2.7
2 1.	6
1 1.	1
1.5	1.6
1.1	0.9

(10 marks)

END